

GATE CSE :

(1) Engineering Mathematics

- Discrete Mathematics : propositional / first Order logic, sets, relations, functions, groups, graphs, combinatorics.
- Linear Algebra : matrices, determinants, eigen values / eigen Vectors
- calculus : limits, continuity, differentiability, maxima / minima, integration
- probability & statistics : mean / median / mode, random Variables, distributions, conditional probability.

(2) Digital Logic:

- Boolean algebra, combinational / sequential circuits, number representations, computer arithmetic

(3) computer Organization & Architecture

- Machine instructions, addressing modes, ALU, data-path, control unit, pipelining, memory hierarchy (cache, main memory, secondary storage), I/O Interface

(4) programming & Data Structures:

- programming in c
- Recursion, arrays, stacks, queues, linked lists, trees, binary search trees, binary heaps, graphs

(5) Algorithms:

- Searching, Sorting, hashing
- Asymptotic complexity, algorithm design (greedy, DP,